Theme Paper on e-Procurement

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Background

The Electronic Procurement is commonly known as e-Procurement which is a part of the information and communications technology revolution that is profoundly transforming the way businesses and people conduct their day-to-day activities. The e-Procurement is a term used to describe many different forms of e-commerce – to add value or reduce the costs of an organization's procurement operations. The e-Procurement is the electronic management of all purchasing activity, often starting with low-value, high-volume, non-strategic goods and services that support the business. It is being used at an accelerating rate by businesses and at all levels of government in most of the developed countries. The e-Government and e-Procurement issues are similar in all nations and each country is on its own journey based on unique geography, people, culture, and resources. Comparatively, the European Union is aggressively engaged in implementing e-Procurement in all levels of government sectors.

A study has revealed that e-Government maturity of any country can be judged on few parameters given below.

i. Use of the Internet for on-line filing of forms & taxes
ii. A government organization that uses the Internet to connect employees, suppliers, and customers
iii. The use of digital technologies to transform government operations in order to improve effectiveness, efficiency, and service delivery

Based on the study report of Accenture Inc (Figure1), Canada, Singapore and USA are the leaders in terms of maturity in the e-Governance implementation, Japan is an emerging performer, and Malaysia and Italy are still platform builders. The developing nations like India, Nepal and other Asian countries are still behind the scene.

The research and study conducted by IT Professional Forum with respect of Nepalese government reveals that most of the ministries, departments, project offices have got their own websites but they are just publishing limited or incomplete information. On the other hand the local government in particular Municipalities seems a step ahead. 30% of Municipalities have informational websites and discharging their obligations towards citizens by performing with some Government to Citizens (G2C) type of e-Government accepting forms for birth, death and marriage registration, citizenship certificate and others online. Some of the government projects supported by INGOs and Public Enterprises, Universities, and Communities have got their websites but most of them are simply disseminating information on their functions, activities and progress.

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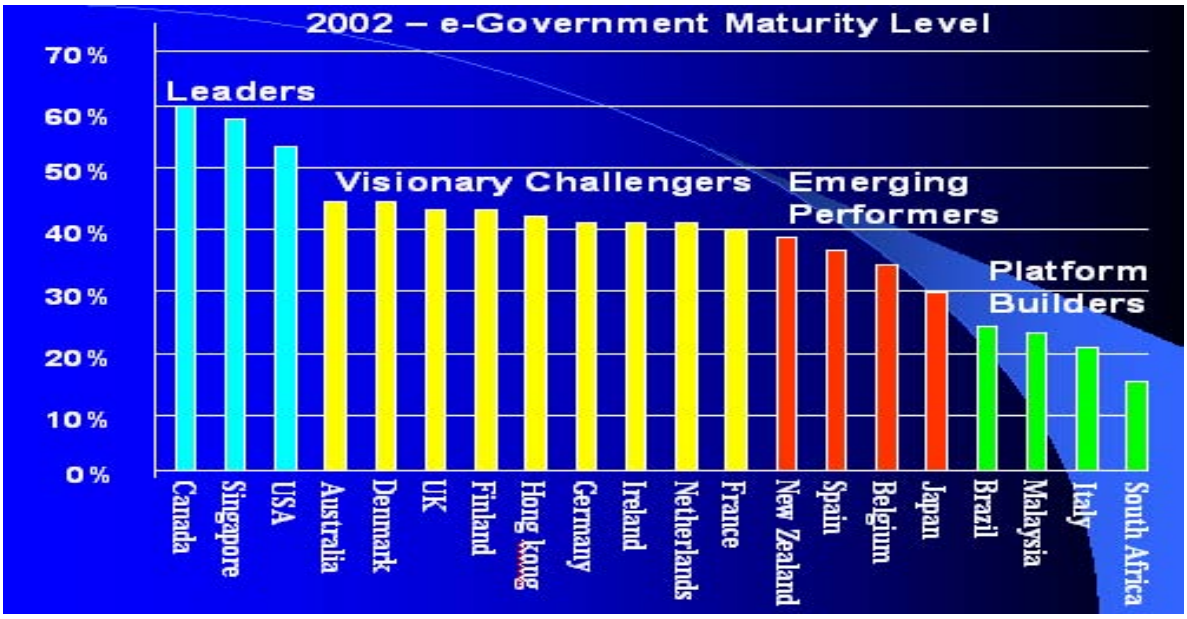

Figure 1 Global E-Government Maturity Level

The global scenario of companies discloses that they spend 36% on an average for external purchase of goods and services. According to Benchmark Research Ltd., 63% of high-performing companies in world wide surveyed in the month of May/June 2002 have saved between 10% and 50% with e-Procurement solutions. The figure 2 clearly depicts the positive result of e-Procurement on price of goods and services, a reduce in cost means a reduction in the sale price and saving at national level.

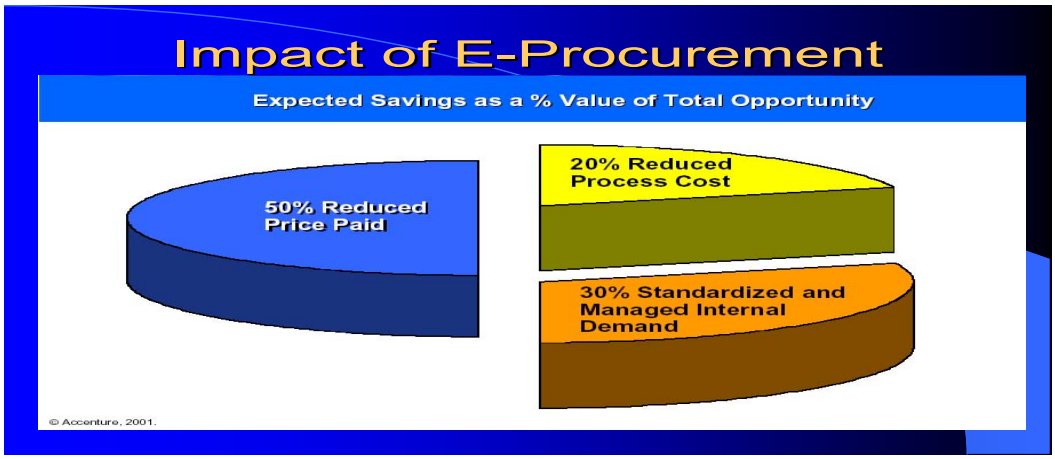

Figure 2 expected savings from e-Procurement

There are many types of electronic procurement models available and being implemented by countries world wide. They are not exclusive rather complementing each other. Depending upon the infrastructure, business demand, market

acceptability and need of a nation, a country can combine different e-Procurement models and are in use today. Let us understand in brief each of the e-Procurement models.

e-Procurement Models in use

The business through Internet is conducted in various ways. Each country is adopting one of the several methods in electronic procurement. It may be useful to begin with a brief overview of all of the four common e-Procurement business models that are widely in use in the globe.

Buy-side e-Procurement

Buy-side e-Procurement refers to a buyer employing electronic systems to purchase goods and services from contracted suppliers and manage all processes relating to those purchases. Purchase transactions are take place over the Internet, and web technology is used to manage the flow of information and transactions throughout the organization and its supply chain.

This e-Procurement system can be deployed throughout an organization, giving staff the ability to quickly and easily raise purchase requisitions and track their progress. The system often incorporates sophisticated workflow technology to streamline the process of authorizing the requisition as well as supplier contract management, which automates the process of transforming requisitions into orders and expediting them with the suppliers. Besides the facilities explained above, it should enforce the use of preferred suppliers and manage many of the financial processes such as the reconciliation of suppliers' invoices, commitment accounting etc. The figure 3 exhibits that a buyer dominates the market and can have more than one supplier supplying different commodities and services.

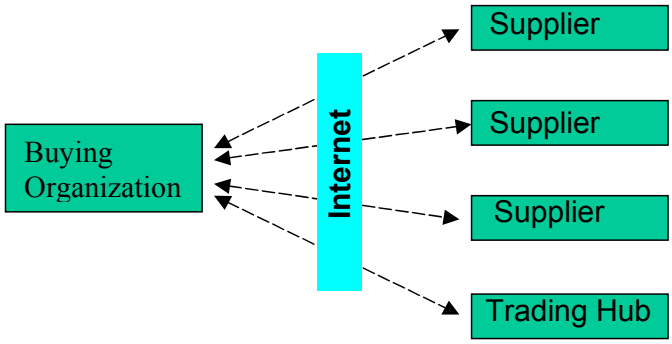

Figure 3 Buy side e-procurement Model

Some of the benefits to the buyers and sellers of Buy-side e-Procurement business solutions are given below.

i. The elimination of administrative costs
ii. Reduced purchase cycle times
iii. Better stock management
iv. Greater company-wide productivity
v. Reduced internal and supplier errors
vi. The ability to manage procurement strategically in accordance with the wider business objectives.

Sell-side e-Procurement

Sell-side e-Procurement describes how one supplier or distributor sells to a number of buying organizations using electronic systems. The Seller may not often use e-commerce technology in selling the goods and services. Although the model is sometimes referred to as e-Procurement but is more correctly be referred to as 'e-sales'. This model does not provide real means of controlling the buying process other than a simplified means of placing a supplier specific purchase order as compared with other models.

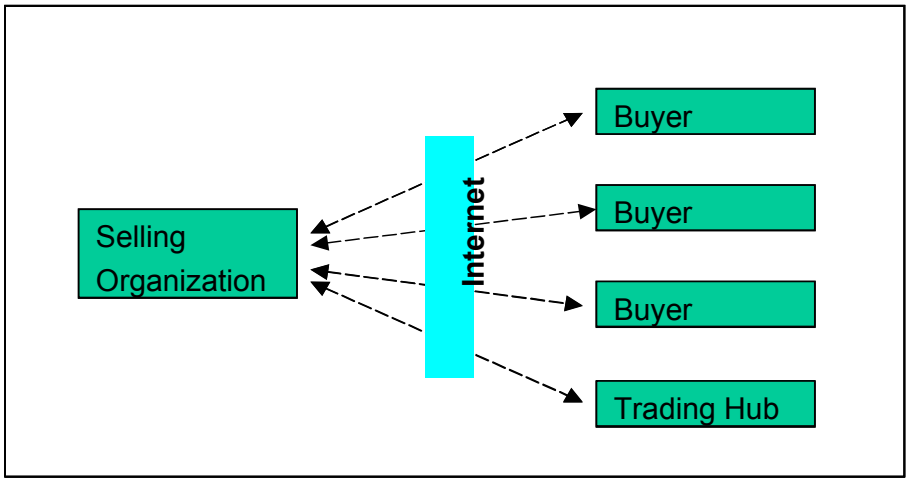

Figure 4 Sell-side e-Procurement Model

The Sell-side-e-Procurement model is driven by the supplier as shown in the figure 4 above – one supplier selling the same goods and services to many buyers. This model is playing a significant part in Business-to-Business (B2B) type of e-commerce. Many suppliers are taking advantage of the low cost of selling over the web and they offer their products at dramatically reduced price, which in turn make the web an attractive place for buyers to shop.

Marketplaces Model

The marketplace model brings together many different buying and selling organizations in one trading community. This model can be compared with the existing common market place where buyers visit sellers' shops and make purchase. It is most popular e-Procurement marketplace and currently several thousand independent marketplaces are in operation in the world and their number is growing day by day.

The marketplace model is most useful in very specific vertical sectors (specific products and services), perhaps with a limited number of buyers and suppliers. The automotive industry is one example of where the marketplace model has been put to good effect, the wine industry is another. Marketplaces defeat the very essence of the Internet because it can not eliminate the similar concept of middleman in traditional selling. Like middlemen, the marketplace must add significant value to the supply chain for it to be worthwhile (figure 5). This model assumes that price is the only major factor in the decision-making process. They prevent the procurement professional from forming any meaningful relationship

with their suppliers, which is often critical to the procurement process in order to be sure of supplier performance in other critical factors such as quality, delivery, and response times and so on.

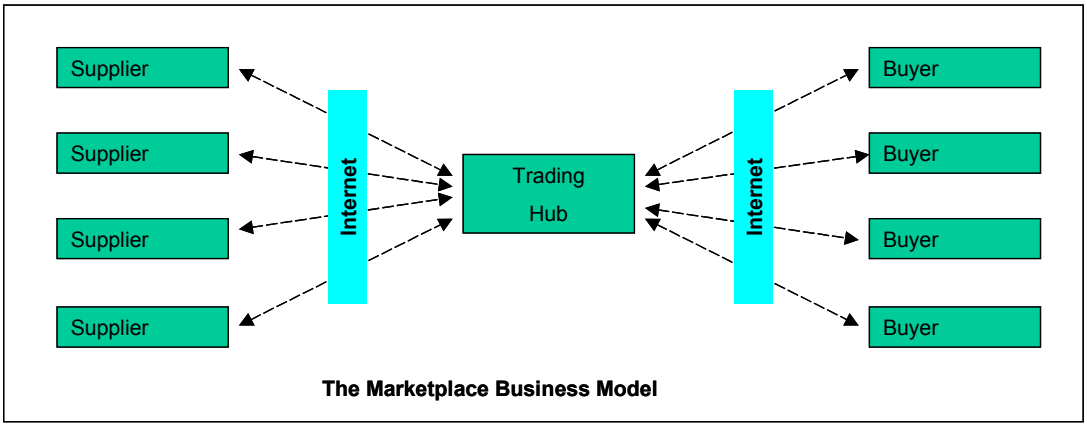

Figure 5 Marketplace Business Model

The Marketplaces do have many benefits to offer but they must not be restrictive and allow other models to join with them. The Marketplace model should be combined with intelligent Buy-side-e-Procurement systems in order to represent a viable business model and add value to the buying process. During 2001, marketplaces were forced to prove their value – this led to a shake-up and many weaker players simply disappeared.

Auctions and Reverse Auctions e-Procurement Model

An auction is a competition among many buyers, each buying organization is try to offer higher price than its competitors. The auction e-Procurement model is commonly used in bidding one type of product such as raw materials, computer equipments, oil & gas and so on and so forth. This is becoming increasingly popular among the procurement professionals who are responsible for buying large volumes of one particular item from the market.

On the other hand a reverse auction is defined in two ways. First, a reverse auction is where the procurement professional auctions off unwanted or surplus stock to other buyers. This represents an excellent means of disposing off what is often a major unnecessary material that is a cost burden to the organizations. Second, a reverse auction is like an Internet Invitation to Tender. It is where the buyer states the requirements using a specialist reverse auction portal or on his own company's Internet or extranet site, and potential suppliers bid for the business.

The auction and reverse auction through Internet is still remaining in their infancy, but they will almost certainly play a major role in Business-to-Business procurement in years to come. Any organization involved in the procurement of large quantities of goods should explore the auction possibilities in their own market sector. Auctions, similar to marketplace model shall be combined with intelligent procurement management systems to be successful.

During the several interactions with government officials and private entrepreneurs, IT Professional Forum explored the possibility of implementing e-Procurement in the government offices. We are detailing the e-Procurement scenario in general and in Nepalese government in particular in the following paragraphs.

Government – the prime buyer

The His Majesty's Government is one of the largest buyers for the procurements of goods, services and works in the country. The state is run by the government through the central secretariats with the large decentralized organizational structures at local levels. The decision making processes is purely on paper-based having complex bureaucratic internal practices. All organizational hierarchy from central to local spend huge amount of money on purchase without proper information on product, price, quality; lack coordination among themselves and loose control over the procurement procedures. In order to overcome the existing paper based problems and inefficiency, e-Procurement is the emerging alternative.

In simple words, e-Procurement is the purchasing of goods and services using the Internet and it covers full life cycle of purchasing (requisition to receipt of goods to making payment). It connects buyers and seller through electronic exchange of catalogs, contracts, purchase orders, Invoices etc. As discussed earlier about the different business models, it includes variety of modus operandi such as Request for Proposals (RFPs), Quotes, Auctions and Reverse Auctions for any procurement. The e-Procurement will able to create central national suppliers and buyers database once it starts. The most important factor for the government and business community is the information dissemination to larger community, whether it is government information to community or the business information disseminated to the government and community.

Some stakeholders consider stock control to be an essential part of an e-Procurement strategy, and the level of procurement related functionality offered by different solutions varies from one to the next. e-Procurement business model will give greater management control and ultimately bring transparency in all stages of government procurement processes.

The activities associated with e-Governance and e-Business within the frame of the government hierarchy is elaborated in Figure 6. The e-Business is divided into Electronic Service Delivery - where the government is delivering her services towards people through electronic medium, and Electronic Procurement – where the government is buying various kinds of products, services through contract, tender, trading and making payment for all purchases.

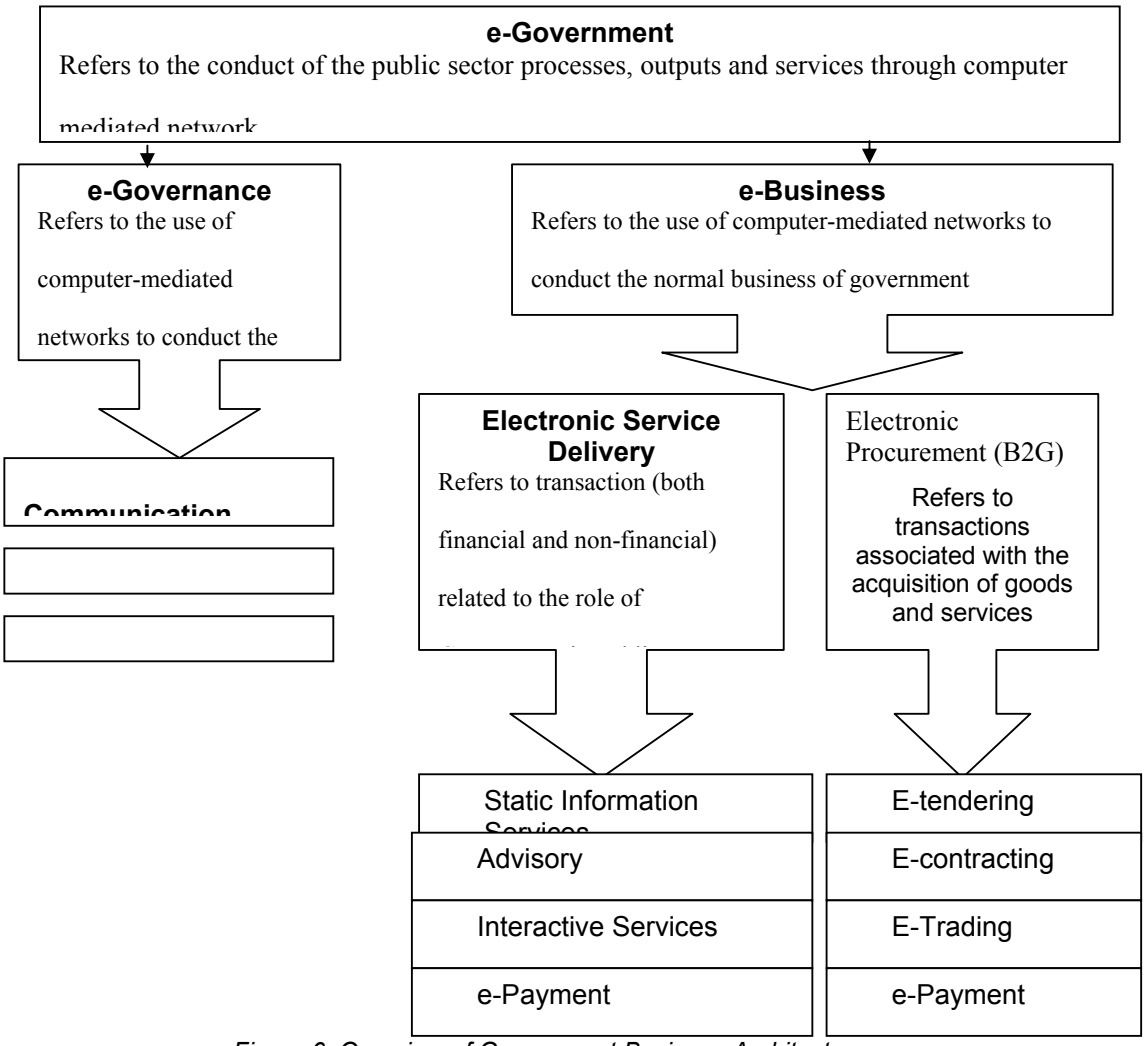

Figure 6 Overview of Government Business Architecture

Procurement model Selection

Selecting the appropriate model for the government is not a big task. In our case, considering the nature of government business of delivery of services to the people and purchase of commodities and services for proper functioning, Marketplace model with Auction and Reverse Auction as explained in the previous section may be the best solution. The relationship of Government-to-Business (G2B) is best known as e-Procurement - the use of on-line technology to enable government agencies to buy or otherwise acquire goods and services. In its ideal form, e-Procurement can streamline the very labor-intensive procedures and processes by which goods and services are sought after negotiated and eventually procured by the government.

The Financial Administration Rules of the country allows the procurement personnel to follow different procedures for buying goods and services depending upon the amount of purchase. Purchases up to rupees one hundred thousands can be made directly from the shop, more than one hundred thousand and unto one million can be made by calling three sealed quotations and over one million can be made by calling tender. Traditionally, government entities have two major

procurement processes in the case of tender; they are Competitive Sealed Bid (CSB) and Competitive Sealed Proposals, also known as Request for Proposal (RFP).
In the Competitive Sealed Bid process, the government publicly solicits sealed bids. The bidding is either open to the general vendors or limited to pre-qualified vendors known to sell a particular product or offer a specific service. The bids are opened and awarded to the lowest and responsive bidder. A responsive bidder is one whose bid conformed to the terms and conditions set out by the government in its solicitation. There is usually little or no room for discretion by the government in making an award pursuant to a Competitive Sealed Bid.

Through the Request for Proposal process, vendors are asked to respond with a proposal equipped with a detailed process or design, and a competitive price. A Request for Proposal differs substantially from a Competitive Sealed Bid in the fact that the government has enormous discretion in making the award. From the time the solicitation is issued and a proposal is received by the government, and prior to any award being made to a qualified vendor, the government is able to hold discussions about the proposals received from bidders, ask for formal presentations where vendors are invited to further explain a proposal, and where vendors are often asked to make a "best and final offer" on and above their initial proposal.

The schematic diagram is drawn which clarifies the process for the government to prepare the tender documents and the suppliers to file their respective bids through Internet (figure 7) below.

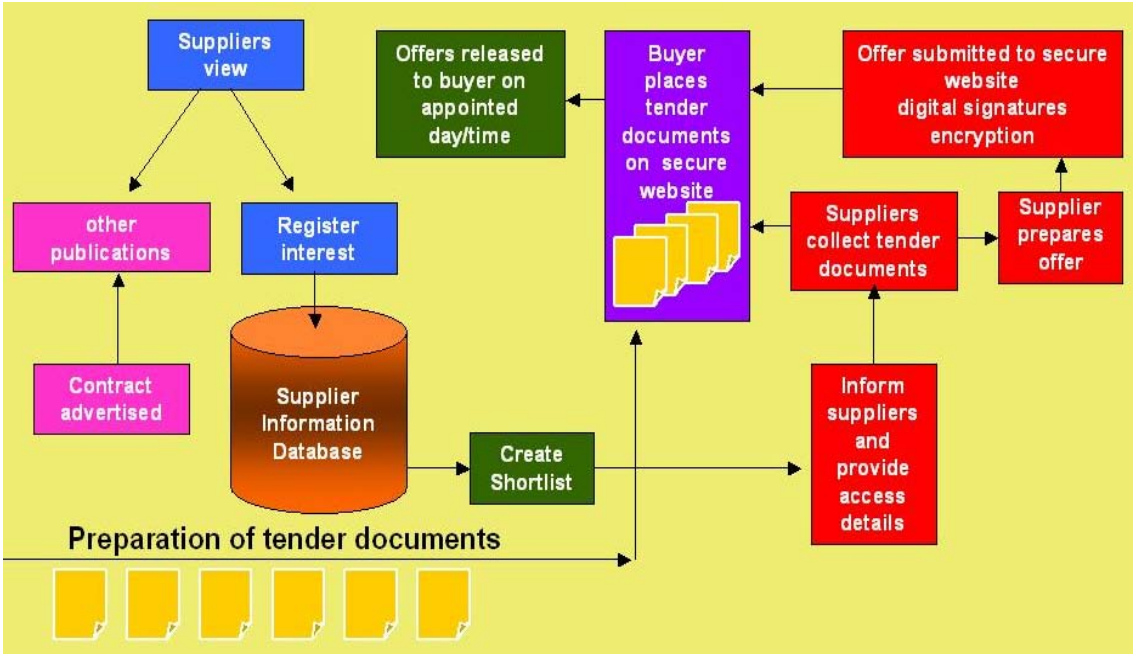

Figure 7 Overview of e-Procurement Process

With the help of technology the government can improve and streamline complicated practices and labor-intensive nature of the procurement and make them simple and more productive. Vendors can cut down various costs such as production, operation, transaction, administration etc by conducting business through Internet and achieve efficiency and pass on the savings to the buyers.

Further they enable buying agencies to find high quality goods at reduced costs in a win-win situation.

Another form of e-Procurement that is taking prominence is the Internet based concept of Reverse Auction with Marketplace model. In Reverse Auction, the government will advertise on-line, either through its own web site or on a third party host site for the need for a particular product or service. A time will be set for vendors to sign onto the site and engage in an auction where they present to the government their lowest bids for the product or service. Vendors are made aware of their competitors' bids and are given the opportunity to amend their bid in order to beat the lowest available price offered to the government. On-line reverse auctions are a fast, easy way for the government to obtain quality products while inducing competition permits the government to achieve the lowest possible bid. Reverse Auctions are most often used in place of competitive sealed bids.

Government-to-Business Marketplace Model

An overview of the Government-to-Business marketplace model is presented in figure 8 below. In the G2B model, there are suppliers within the market place and outside the market place and they are generally registered in the government agencies as a short listed service providers or vendors at the same time the buyers' profile will also be in the market place. The tender is published and the message is send across all the suppliers who are registered for such category of supplies. The process of bidding continued until the contract is awarded and the total business process is completed. The financial management information system just informs the ordered of the pay to the transactional bank. And bank of the supplier will be paid. And the supplier will give the statement about the payment. Here the payment mechanism is shown in depth because it is one of the problem in both traditional and electronic purchase system.

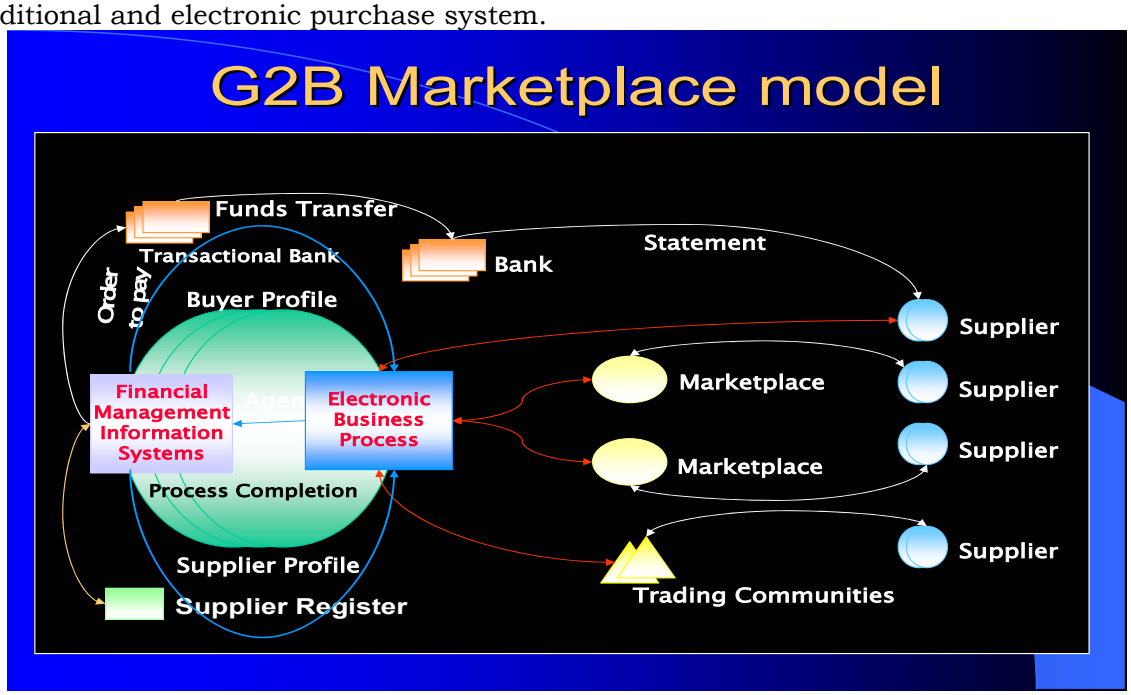

Figure 8 Governments to Business Marketplace Model

Accordingly, before moving down the e-Procurement route government needs to have a clear understanding of how to introduce it in a way that positively contributes to both their business objectives and the expectations of their local communities. A critical requirement for the government is to understand the business processes and technical issues they will need to deal with if they move to online purchasing. e-Procurement can involve much more than just an electronic interface between a buyer and sellers. The whole purchasing and inventory management process needs to be analyzed to maximize the opportunities made possible by the new purchasing process and evaluate the e-Procurement value to government procurement. The figure 9 exhibits the procurement activities compared against traditional purchase issues and value added by e-Procurement.

Procurement Activity	Traditional Procurement Issues	eProcurement Model
Product Research	Slow and laborious process involving catalogs, phone calls, multiple web sites	**All product info in one place, info is real-time and accurate**
Approval	Time-consuming process to document multiple quotes, long lead time for bids	**Audit trail, approval history, automated approvals**
Purchase	Time-intensive, redundant data entry, high processing cost	**Bids, quotes, POs automated. Stop "reinventing the wheel"**
Technology	Installation, training and updates can be expensive or time-consuming	**Browser is all that is needed, updates are automatic**

Figure 9 e-Procurement Values to Government Procurement

Marketplace Deliverables

The e-Procurement system generates much information which is necessary for record keeping & auditing, follows up and provides tools for planning, analysis and monitoring. Initially, the e-Procurement marketplace system will serve as the official system to advertise and distribute specifications for public bidding opportunities by the government of Nepal and other Government Agencies. The e-Procurement marketplace system also contains information related to doing business with the government such as rules and regulations, press releases, a directory of government agencies and contacts, planned and historical agency procurement, bid matching for suppliers, potential competitors, winning bidders, and a frequently asked questions section. In the future the system will be extended to support other aspects of the procurement process including direct purchases, bid submissions, central accreditation and payments.

The pilot design should facilitate better communication, greater transparency, and better supply chain management. In addition, the platform should be expandable, with the potential to become the single portal for doing business with the government. Some of the services that can be available to buyers, suppliers and general public are given below.

- Public access through Internet
- One time registration of Buyers and Suppliers
- Tender Announcement, Download Documents, Drawings & Award
- Electronic Bidding & Quoting
- Advance Notification of contract and opportunities to Suppliers
- Secured transaction
- Payment gateway
- Public and Private Messaging system
- Search matching contracts and offers
- Access to Price Reference for Assessment of Procurement
- Supplies & Service Catalogue
- Government policies, Laws relating to Procurement and other information and notifications
- Total business tool with Contract management and other back-office management solution

Besides the services delivered by the marketplace as explained above there are many benefits to the buyer and seller and they are described below.

Benefits

The e-Procurement will enhance transparency in government procurement since all transactions are recorded, accounted and provided online. Full information of the bid, evaluation criteria & procedure, winning bidder, reason for the award and the contract amount will all be accessible through the e-Procurement Marketplace System. In addition, the electronic catalog, which provides information on the pre-approved cost of commonly used items, will help government accountants and auditors check that supplies purchased by a government agency are not grossly over-priced. E-procurement offers a very quick and simple means of raising a requisition using an electronic form, which is then automatically routed to the various authorizers and then either to the procurement department for conversion into an order, the stores for conversion into a stock issue, or directly to the supplier where a preferred supplier exists and a contract is in place. In many medium to large organizations, the saved time and effort can equate to thousands of man-hours every year. An e-procurement solution stores all data in one central database and enables automated matching and simple mismatch reconciliation. No paperwork is involved and a supplier's invoice can be matched and authorized in minutes, rather than weeks. A proper e-procurement solution will automate the accruals reporting process as well as virtually any other information needs arising from procurement, freeing the buyers to focus on procurement and providing finance with as much analysis and forecasting as required.

Function	Traditional Process	E-procurement
Requisition Generation	65.77	29.2
Requisition Distribution	6.05	0.0
Order Generation	9.87	1.5
Order Distribution	0.87	0.0
Expediting	0.91	0.3
Goods Receipt	3.83	1.5
Invoice Processing	9.40	0.7
Material	3.31	0.0
TOTAL	**100.0**	**33.2**

The above table, taken from a report by Warwick Business School, highlights the administrative costs of a traditional procurement system compared with the costs of e-procurement *(using the manual system costs as the base index (= 100))*

Government

Some of the significant benefits are enumerated as follows.

- ✓ Improved and online database
- ✓ Elimination of Administrative costs
- ✓ Significant Cost reduction in purchase process
- ✓ Reduced Errors vs. Phone, fax, mail
- ✓ Reduced order cycle time
- ✓ Real-time order status
- ✓ Increased information on suppliers and buyers
- ✓ Effective Monitoring
- ✓ Improved Contract Compliance
- ✓ Reduced prices through increased Purchasing leverage and aggregation of demand
- ✓ Proper procurement planning and Lower stock costs

Suppliers

- ✓ Enhance Market Visibility & Access
- ✓ Publish Once & Sell to Many
- ✓ Increases Sales Productivity with 24 hours and 7 days Digital Sales Agents
- ✓ Reduces Administrative Costs & transaction costs
- ✓ Fast receipt of orders
- ✓ Increased accuracy of orders
- ✓ Real-time order status information
- ✓ Automated Order Management
- ✓ Reduced time for receiving payment
- ✓ Decreased Operating cost

Security – a must

It is quite easier for Government to conduct purchase electronically through Internet. But there is a biggest problem in doing business online. Every piece of information is important from buyer's as well as seller's point of view. Neither buyer nor seller wants to disclose the information to any person other than the one for whom the information has been sent. As such security is the major concern for any one doing business over the Internet and this applies in Nepal as well. The e-Government and all electronic business transactions require trust that buyers, sellers and public can accept that they are working in a private, confidential and most secure environment. The security is a primary concern and is a must for carrying out business through net.

The information and communication technology has developed various methodologies for securing electronic transactions. Among them Digital signatures provide strong authentication while Encryption protects integrity of content against interception. In order to consider confidence of doing business online, four major things as given below, need to be secured

Privacy
The e-Procurement model must assure that privacy is maintained while exchanging documents between buyer and Supplier across the web. No other person(s) can get access to them.

Authenticity
The system must promise the identity and authenticity of the sender and recipient of documents and payments.

Integrity
The system must guarantee on the integrity of the document content when exchanged across the web.

Non- repudiation
Neither the buyer nor the supplier should in a position to deny that they had not delivered or make payment.

The figure 10 illustrates the different types of securities that are required to build and operate e-Procurement system in a country. Some security measures are protecting the physical materials while others protect the network, logical component and databases.

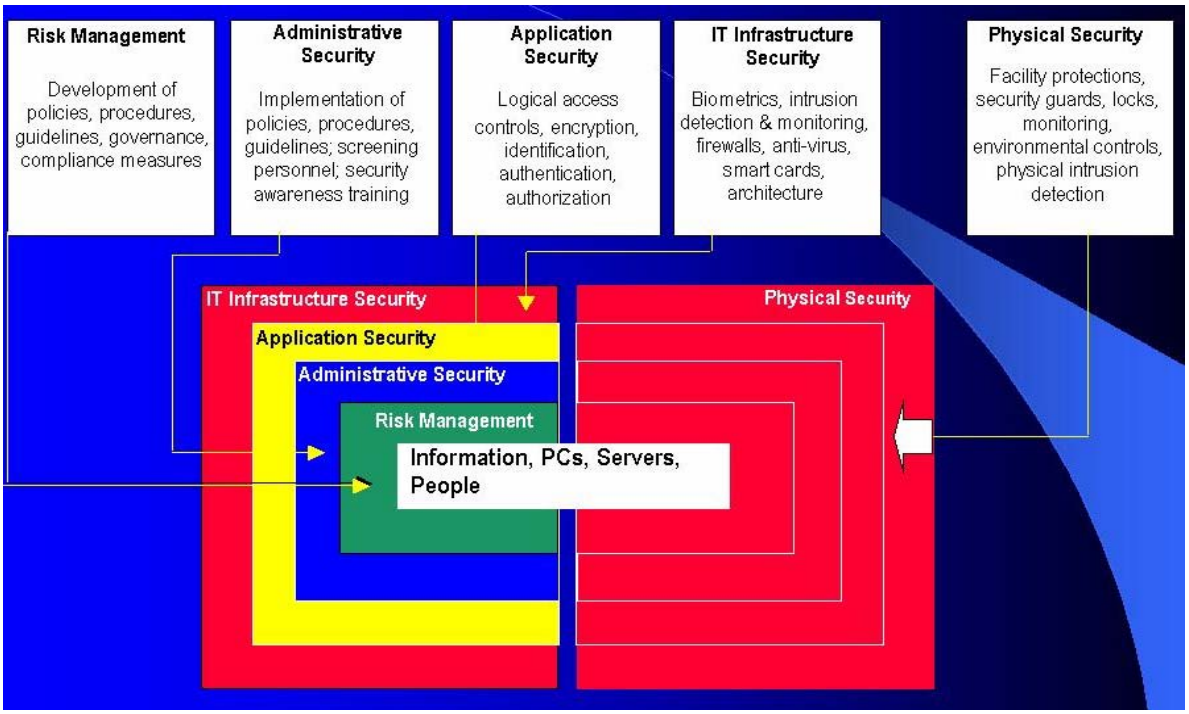

Figure 10 Security Overview

Challenges and Opportunities

The e-Procurement is a very new concept developing in Nepal and might take sometime before it gets mature. Of course there are many challenges ahead in the process of building, operating and implementation of e-Procurement. The successful outcome will improve the country's competitiveness, open new avenues and opportunities for public and private within and outside the country and its failure may set back developmental activities. Hence the challenges should very vigilantly manage so that the success should appear in time.

There are many challenges ahead for the success implementation e-Procurement in His Majesty's Government of Nepal. It requires strong leadership and existing top-down approach is critical for this. Besides that there are many issues like difficulty in recruiting, retaining, and training of in-demand technical and business skills, digital divide in all levels, extremely large pool of heterogeneous data and fragmented service delivery system. Traditional budget process makes funding of cross agency projects difficult. Another major challenge is the commitment to the business model and approach chosen to roll-out. None of these components necessarily require the automating of anything but involve the detailed scrutiny of the way business is currently conducted and the things that may need to be considered in terms of moving towards an automated system. The important issue is to focus on procurement process improvement before automating anything. The utmost considerable challenges for the success of e-Procurement are as follows.

- Organizational support
- Supplier Management
- Integration to Legacy Systems
- Security and data sharing
- Buyer adoption

How does Nepal get there?

For the country like Nepal, where most of the global achievements are introduced much delayed, we can think of implementing e-Procurement in phases only. The proper infrastructure development and planning the roadmap of activities during and after transition period may take to the destination without loss of time and resources. The country needs to accommodate many changes at various places and build new approaches as well. Some of the fundamental changeovers are described hereunder.

It is difficult at this stage to identify the transition period but a careful planning for transformation to the e-Procurement relies heavily on the use of the Internet and other electronic methods to communicate and deliver services by government offices. One of the purposes of e-Procurement is to streamline the purchasing flow and associated approvals. Government Agencies should take this opportunity to re-evaluate and update their business processes and streamline their approval procedures.

Many commodities are similar in look and size but they may differ in quality and price. The buyer could not get access to the commodities physically while selecting through Internet, it is very important for manufacturer or suppliers or buyers to identify the specific commodity by its identification. It is one of the biggest tasks before the manufactures and suppliers and such codes should also be brought to the knowledge of the buyers before we move towards e-Procurement. Thus, codification of materials should be standardized for proper product identification in marketplace. Of the many kinds of marketplace model account for in previous sections the government should define, identify and build the right model in the country by public or Private or Public Private Partnership.

It has been important for the government to understand that the e-Procurement Marketplace System is a business solution and not simply a technology solution. Increased efficiency and business transactions will not materialize with the establishment of the e-Procurement Marketplace System if the government does not improve its overall operations -- including delivery time, quality of products, cost reduction and accounting procedures. Initial steps should be taken to address these issues, however, considerable resources and efforts are still needed to move the system forward. As an infrastructure to the Internet based business in Nepal, there are few concerns, which should be urgently addressed, worked out and implemented:

- Ratification of Digital Signature and e-Procurement Act and amendments in many other related acts
- Government should prepare Security Infrastructure plan
- Inter-connectivity among government agencies
- Considerably high speed Broad-band internet availability
- Payment Gateway and Payment Cards issuance
- Marketplace development and Content development plan in National languages and English language
- Online marketplace Monitoring and Dispute Management plan
- Training for government and suppliers on new culture of doing business
- Incentive plans for Vendors using online method of doing business

The Government commitment is very important for implementing e-Procurement at all levels of its operations. In order to establish and operate e-Procurement in the government ministries, department, and offices and at local levels the government needs to approve the vision, strategy and set objectives and define plan of actions. We have trying to illustrate on these aspects in the following sections and will be a guiding line for government to draw a realistic approach for the implementation of e-Procurement.

e-Procurement: Vision

1. For the employees in government agencies, e-Procurement must be easy, secure and user friendly. The system will need to integrate with the agency's other systems to enable 'real time' financial and transaction management of procurement expenditure. The system should be made available to all agency employees with procurement responsibilities, and should be accessed using internet/intranet technology on any PC/Thin Client in their place of work. The 'user' should have controlled access to the system on the basis of personal password.

2. The content of the system should be based on user requirements and need to be flexible enough to allow for emergency purchasing and 'spot' purchases where appropriate. National language and English languages can be used for the content development.

3. Where it is possible, the content should appear as one catalogue to the user and they will select the items/services they wish to purchase directly from the screen with details of the product/services and conditions. The user should be notified of the availability and delivery date and the supplier/contractor will then meet the order directly.

4. All procurement through the Corporate System should be subject to the appropriate consultation with the Agency's Internal Audit. This would include checks to ensure compliance with government legislation; Financial Regulations, Best Value Procurement Strategy & Policy and protection of the Agency's interest by the use of appropriate terms and conditions.

e-Procurement: Strategy

Lack of transparency in government procurement has been a systemic problem in Nepal. In addition, inefficiency due to manual and paper-driven procurement processes has resulted in poor service and low sales volume. The study by IT Professional Forum found that establishing an e-Procurement System could help improve transparency, efficiency, and value for money in government procurement. Stakeholders (government agencies and suppliers) showed their support for the e-Procurement concept. The government should capitalize this concept and encourage community to go in the direction of e-Procurement.

The strategy should recognize that there are elements of risk, which are unavoidable. If the full benefits of e-Procurement are to be realized it should be managed very cautiously. The implementation of the e-Procurement Strategy will face a number of potential barriers and challenges. To help the change processes take place, key areas of concern have been identified and ways to address those potential issues noted.

The Action Plan relating to the e-Procurement Strategy is to be developed and continually reviewed in light of the progress and developments in information communication technology and the potential future issues within government agencies and suppliers.

The expectations of the e-Procurement Strategy must be approached with extreme care, otherwise a great deal of time and resources could be committed for very little gain. The agency's options and opportunities for e-Procurement must be kept realistic, it must focus on practical and achievable options that will give the added value in terms of increased efficiency and effectiveness whilst minimizing the risk to the agencies.

e-Procurement: Objectives

1. The intention of the Agency should be, to make the best use of information communication and technology to reduce the cost of the procurement process for both the agencies and its suppliers and contractors.

2. The areas of anticipated savings/benefits relate to the followings
   - 2.1 Reduced procurement costs
   - 2.2 Consolidation of buying/purchasing power;
   - 2.3 Reduced 'off contract' purchasing;
   - 2.4 Automated transaction process thereby releasing employee time;
   - 2.5 Increased control of corporate spending (provides real time commitment against budget or available funds. Also instant updating of financial reports);
   - 2.6 Improved management information;
   - 2.7 Improved key supplier relationship;
   - 2.8 Improved supply chain management;
   - 2.9 Reduction in 'manual' invoice volumes;
   - 2.10 Replacement of paper based purchase orders;
   - 2.11 Reduced central support services costs;

3. The prime cost savings would derive from the followings
   - 3.1 Reduced procurement costs;
   - 3.2 Employee salaries;
   - 3.3 Telephone call charges e.g. orders, quotes, queries etc;
   - 3.4 Purchase order delivery e.g. stationery, postage stamps or fax costs;
   - 3.5 Capita charges per invoice processed;
   - 3.6 Bank charges per check payment.

4. The main efficiencies would derive from:
   - 4.1 Online procurement;
   - 4.2 Employee time;
   - 4.3 Speed of the procurement process.

5. The Government Agency needs to measure the success of the e-Procurement strategy in tangible terms. This means setting Key Performance Indicators.

Key to Success

The implementation of the e-Procurement will face a number of potential barriers and challenges from within the government organization to outside world. At the

outset, it is important to recognize that resistance will be elevated from among bureaucrats and they will be quite slower to accept change. To help the change process take place, other key areas of concern have been identified as below, where the government should focus from the planning phase.

- Supplier adoption – There is a need to get the suppliers on board. Not all suppliers have access to PC's or web-enabled.
- Supplier not ready for Purchase Cards;
- Corporate Agency Adoption – e-Procurement must feature in the Agency's Corporate and Service Development Plans.
- Change Management – Key to the success is training, communication and resources.
- Technical Integration – There is still a lack of universal standards (though getting nearer) for e-Procurement systems.
- Culture within the Agency – A team approach is required and a 'Can Do' attitude must underpin the program.
- Legal issues around accepting tender electronically – for example, electronic signatures.
- The extent of savings achievable through e-Procurement.
- Impact of e-Procurement on SME's and local economy.

However a unified effort of all stakeholders will avert the failure by addressing the above issues timely and promoting the following activities promptly.

- ✓ Actively involve all major stakeholders engaged in e-Procurement;
- ✓ Actively collaborate with suppliers, partners and purchasing consortia;
- ✓ Processes must be 're – engineered' not simply replicated;
- ✓ All Service Units and systems must be included (not just selective parts);
- ✓ Make e-Procurement 'evolutionary' not 'revolutionary';
- ✓ Implementation must be continuous;
- ✓ Keep it simple – avoid trying to do too much too soon.

The safest approach to adopting a new technology with a steep learning and adoption curve is to take small steps with activities that are manageable within relatively short timeframe. This allows greater flexibility for tailor-fitting the system and formulating a long-term strategy based on the actual experience of the organization and feedback from the clients. When it comes to choosing a technology or system, there is no need to reinvent the wheel. Customization of an existing system is often much easier. It is very important to know that e-Procurement without an underlying foundation of strategic sourcing only lets you buy the wrong thing from the wrong supplier at the wrong price at lightning speed. Therefore it can be demonstrated that the government should take a careful approach in making decision to go ahead with electronic procurement system in Nepal.

Marketplace System Requirements

We are presenting below the potential system requirements for e-Procurement, once the government decides to build and operate e-Procurement marketplace system in Nepal. This is not the exhaustive list rather a brief one to give fair idea what the basic needs are.

1. Security user log-on and password should be necessary once only in order to access the e-Procurement system, suppliers and any approved purchasing method.
2. Easy to use system where employees can select supplies and services from on-line catalogues, place orders directly with suppliers, and with integrated approval by budget holders.
3. Seamless ability to exchange business documents with suppliers (including quotes, tenders, purchase orders, goods receipt notes invoices).
4. Provide maximum use with the minimum training.
5. Access should be controlled to those suppliers that have been previously approved, and hence prevent 'off-market' purchasing.
6. Ability to take a single view of multiple suppliers and their catalogues for comparing products and prices.
7. The system should be comprehensive, fully integrated, and scaleable with potentially unlimited users and be modular to enable phased implementation.
8. The system must be able to deal with multi-currency.
9. The system must be secure and provide comprehensive audit trails.
10. During the potential implementation phase, and adoption by suppliers, it should be possible to print out documentation for transmission by FAX to suppliers that are not able to receive them directly from the e-Procurement system.
11. Government should set up a Customer Service Area and create an E-Procurement Marketplace System Division to assist and answer queries from government agencies and suppliers. Some public terminals should be set up to provide access to E-Procurement Marketplace System Division for those who do not have a computer or Internet connection. Still, the issue of accessibility will continue to be a challenge in broadening the base for the E-Procurement Marketplace System Division.

A strong e-Procurement organization should be formed or designated to one of the organization which should be the highest governing body for government procurement. This organization will need involve with the e-Procurement Marketplace System long-term planning & policy designing, deploying human resources, infrastructure and technology, framing rules and regulations, managing the marketplace, bringing awareness' and conducting different levels of trainings and encouraging implementation to ensure the system becomes the central portal for government procurement.

Pilot project

In order to assist government agencies work through e-Procurement issues, a pilot project is scoped by IT Professional Forum. Upon the decision of the government, the e-Procurement project office shall be planned, designed, implemented and monitored and its success will be replicated to other government offices in due course of time. The pilot project will test all the hypothesis, infrastructure, security and network constraints as well as facilities and will produce actual result which enable all stakeholders to put forward the resources for using information and communication technology in the economic development of the country.

Reference

1. *The Constitution of Nepal 1990*
2. *Arthika Prasashan Niyamavali 2056 amended in 2058 B.S.*
3. *Local Self Governance Act 2055*
4. *Public Procurement Act 2059 (draft)*
5. *Country Procurement Assessment Report by World Bank, April 2002*
6. *Financial Accountability in Nepal – A Country Assessment by World Bank, March 2003*
7. *Commerce Outlook 2060 – Published by HMG of Nepal, Department of Commerce*
8. *IT Policy 2000 – HMG of Nepal, Ministry of Science and Technology, 2000*
9. *Policy Recommendation on Electronic Transaction and Digital Signature Act – A Study Report by IT Professional Forum Study, December 2002.*
10. *www.bharatpurmun.org.np*
11. *www.Gem.wa.gov.au*
12. *www.worldbank.org/publicsector/egov*
13. *www.comprasnet.gov.br*
14. *www.compraschile.cl*
15. *www.ehandel.no*
16. *www.gebiz.gov.sg*
17. *www.palermoconference2002.org*
18. *www.procurementservice.org*
19. *www.unpan.org/egovernment2.asp#survey*
20. *www.unpan.org/e-government/global%20leaders%20index.htm*
21. *www.e-government.govt.nz*
22. *www.eprimers.org*
23. http://www.unpan.org/e-government/global%20leaders%20index.htm
24. *Central Bureau of Statistics*
25. *A Study of Provisions & Practices of Accountability in Nepal Civil Society by Transparency International Nepal, August 2001*
26. *'Foreign Assistance and its Audit' – A paper presented by Rt. Hon. Bishnu Bahadur K.C., Auditor General in a workshop on Transparency in Development Aid in 2001*